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Hello guys, how are you? Great, good, thank you. How are you? Yes, yes, I'm good too. Great.

Sorry I'm a bit late. No problem. Sorry, I was on something else.

Okay, great. Maybe I just want to start with two things that I did, so I'm going to be the student. But no, there's one point of conversation that I want to open while Jean-Philippe is here.

But just hold on and we'll share that screen. So, I couldn't wait and try every image model that I could find. So, I tried Gemini and Chad GPT 5.6 model and also Cloud.

Cloud doesn't have an image model, but you can use tool to extrapolate. So, at that moment, Gemini and OpenAI completely failed. So, I told them just try to extend the frame of that image.

Chad GPT some time. So, I'll just show you some of that. So, that was the original frame.

So, this one was really like a spherical contact point here. The object was a little bit squishy. So, this is the reason.

So, you see the original frame. So, if you would see the original image that I gave, it's mostly that real data. And you did understand that there were nothing to extrapolate beside that part here that go a bit away.

So, it would be pretty functional to work with this one. I did this one, which was kind of a, I grab a cubit or like a, and you can see like that was the edge of the object. So, the edge is pretty clear and asking to extend it.

So, this one comes from Cloud or Open? Yeah, Cloud. Wow, nice. Yeah, actually, he worked this out as a matrix of color between 0 to 255 and then just ask it to display with dot.

So, sorry, it's just a bit exciting. It looks like he used like just basic model to extrapolate. He doesn't take any like knowledge priors or any consideration of possible shapes or anything.

Just like the way, it's like a flow, thermodynamic flow almost like, I don't know. I feel he's really doing like, personally, like when you say thermodynamics, it's like taking the gradient. Personally, it's the infinite element and you just try to extrapolate with the gradient.

And so, yeah, that's this for this contact and that was with like a shape like that. I think it was a sharpie pen there. And the point is, okay, the good thing is the best model was Cloud and he wasn't using image model.

So, that could be pretty fast, okay, because at the moment, real image model take like almost a minute to compute. So, when I ask chatGPT or Gemini to update like the frame, it could take a minute. So, there's no way to speed that up.

The fact that it's not, it's more like a regular request where you do some math. Yeah, I think it could be applicable. But the thing is right now, it's like you said, there's no knowledge of the sensor.

And I think it's where even though this would be the best and it's kind of not too bad. You see here like you would probably understand like a real image model if we train it would have understand that usually like the textile is not as strong because there's a bit of curvature on the sensor. So, when you have a rigid object, you might not see a lot of simulation, but it doesn't mean that the object does not continue.

So, I'm sure it's kind of sensor behavior that the machine learning would capture. So, that was just to update you. These are really interesting.

I mean, how did you, what was your prompt to, it gives you back the matrix? Yeah, I can send it to you, it's saved somewhere. But yeah, it was like a prompt like here's like a tactile image of a matrix of that size, that size. You see the BQB conversion of it, but you see the textile and then try to extrapolate like which would be the contact outside of that, like assuming that the object can be bigger than that.

But there's one point, JP, that I wanted to talk with you. So, we thought about like it was mostly a problem of image stitching. Yeah, yeah.

It's where I realized it is not as simple as that for some case, and I will explain why. So, okay, let's say I grab that ball. So, the point is image stitching is usually you have like for the microscope, for example, you have the full image, like the full depth image of every place.

And then when you move, it's always the same one, but you stitch it somewhere differently. And that's kind of true for the ball, for example. But if I do the ball, let's say that's my tactile sensor.

So, if I move it, it's always the same contact. But as I will pass the maximum height of the point, I will start to make appear this contact at the tips. Yeah, okay, yeah.

Which will not be captured. So, the point is as long as I stay in the same point of contact, since the point is with the sensor, you only get one layer of contact. You don't get like the full depth map of the contact.

So, the point is you only see the contact you get. So, like you have this image, you have this image, you know you move. So, it's easy to stitch to just you know how much you move.

But as you start to go and shift the point of contact, it creates a formation you haven't seen before. If it would be a vision-based image, you would have seen that point here. Like the contact with the side of the ball here.

Yeah. But right now, you will never have seen it. So, it's kind of like it's going to appear a point where there were nothing to stitch before.

Yeah, but I would still call it stitching. It's just like I don't see how… Okay, suppose like let's talk in… Like let's consider that the tactile map is a deformation map or that we cannot process it into a deformation map. And that we always reference like this deformation map in an inertial frame.

So, that when you… The example you were providing where you have like a sphere touching a different location on the sensor. It's just redundant data actually because you're always touching the same point on the sphere. And then as you exceed a little bit the center of the sphere, you're starting to discover new region.

Then the deformation map is just… It will generate new points in the inertial reference frame. But okay, so what you're saying is like that there's no overlap in the previous image. Like if you recenter in the original frame, it's like assuming that you have a bigger contact.

Oh, okay. If you stitch it in the original, like you move from that position. Now, like the original point is outside of the frame, but now you still see one in a frame.

Okay. So, if you recenter that is… So, it's kind of like if you do that, you have to create a layer of… If you cannot stitch it and there's new things that appear, I think we would have two layers. You have to assume that you are in a deeper layer.

And I think if we can work it out, it's actually one of the limitations of the initial method. Because when we're thinking of this spherical contact, we're just moving that dot right now where there's other dots that exist. But one thing that you're totally right is that for the stitching algorithm to work, at least the one I know, they need to have a bit of overlap all the time.

So, if you move too far away, even if it's a sphere, theoretically, I know it's like one point of contact. But in reality, the sensor will… Since it's soft, it could adapt a little bit to the shape. So, if you do a tiny movement, then you will be on the edge of the sensor.

But you can maybe still have a bit of overlap in the deformation map and then proceed to stitching using this overlap. But I'm not sure. I'm not sure.

I get your point. To me, it's whenever you see new information appearing like that, it should probably go into like an RGB image on a deeper layer. And then just think about a part where you would actually have like an RGB image.

But you know RGB is just three colors stacked. Yeah. It could be like kind of a depth image where you have like the depth information.

So, we just indent it. So, let's say you have a part. So, do I still share my screen? Yeah.

This one. So, let's say you grab that plus shape part again and you see the plus by the side. So, first you grab the top of the plus sign and then you move, you move, you move, you move.

And then the plus side, the plus top edge goes out of the image. And then you close on that new depth further down. So, you generate an image, some contact that would not fit with the original one.

If you stitch it to the first layer, it creates issue. Yes, yes. That layer don't exist outside of the, like inside of the original frame.

It just exists outside of that frame. So, point is, it tells you that you're deeper now. So, it's an information you couldn't see before.

Yeah, but to me it's not, correct me maybe I didn't get your example right. But in this case, the problem is that you have no overlap at all. It's new information.

You cannot try to like stitch it to previously acquired. Well, I mean, it's a digital example. I understand that there's no, but you could probably still stitch it because you know exactly where you are.

So, you have new contact information at that point. But the point is, like if you stitch it, like if you totally understand that that contact goes and you can reimagine the object with it. If you just stitch it to the original point, where that new contact would go.

I'm not sure I get what you mean. So, you were saying for the plus sign, you grasp it by the side and then you start grasping like the extremity of one edge. First, you have like a rectangular contact with the upper edge, both upper edge of the plus.

So, and then you go out of the plus and then you grab the side part of the plus shape. Yeah. But the point is, if you just stitch it because you know the reference frame you are.

If you go back and stitch it to the original part, you will stitch part that don't exist. You will stitch part. But in this case, sorry, maybe I'm too much.

But in this case, I would say we would just have to register the new tactile data in the reference frame without stitching because we figured out that there were no overlap between this and the previously acquired tactile maps. But to me, the main point I'm making is as long as, so when you stitch like stitching algorithm, have all the information. They might not record the depth, but like an image, you would still see that part.

All frame would see that part. Let's say if it's in depth, you would see it more like in gray instead of very strong black or I don't know. But the point is, you could just teach it.

If you move the first part of the plus outside, you still see that one and you stitch it at the right place. But the point is, that reference contact layer doesn't exist around the center. But it's a bit like a mobile robot exploring like a completely new room.

The mobile robot has a 3D LiDAR, suppose. And as it moves in a continuous room, let's say, it can stitch the point clouds. But suppose suddenly it passed through a door and it's a completely new room.

There's no overlap. Then we can't stitch. So we can just register.

The main point is you have depth in that case. So it's pretty easy. It's new point.

And you put it at the right place. Where would you put that? For me, depth would be like a tactile imprint in the case of a tactile map. I just want to come back.

Let's say that's the sensor. It's a big deal. You have that contact.

It's like a Japanese flag. And as you move, you see that point moving. You took all the data.

But since you have the reference frame at the end, you know exactly that was the contact. The way we assume it right now is going out of the contact just means we have nothing else here. But the point that what will happen is when we will start to lose that contact, we will have a new circle of contact which is going to be bigger.

So when you stitch it back, it's okay. You know the reference frame. But it's kind of like you say, okay, well, in that case, the initial contact was the big one.

How do you handle the big one? Like I said, when you have the depth, in your example of the litter, it's okay. You put the new point where it is, the new information where it is relative in all the axis. Now, since you just have a view on a plane, you can only see what happened on that plane.

But when the object due to convexity starts to go out the initial contact, it created a new contact that didn't exist before. That layer of contact cannot be reconciled with this one. If you recenter that contact, because you know, you move the frame from that centimeter.

So the original contact is out, but you have a new one that didn't exist. And in reality, in my example with the plus sign, it's going to be very binary. So you would see like a new contact appearing whether or not.

So in my example, the plus sign, you would have seen like a rectangle all along. And as you move out, you would discover like a wider rectangle that would like go like that. So how do you stitch it if you're just on a single plane? Because if you reconcile them on that plane, around the middle, in that plane, in the same plane, those two contact points don't exist.

And again, the really big difference with stitching like a LIDAR stitching image like in a microscope is we only have the view of where we touch. And where we touch depends on where we are in the object. So we don't have like the full image.

We just have like one layer of image. And that layer will change as we will move outside of the grasp. And it's probably something like if we deal it right, it's better.

It's actually an improvement of what we used to do because the way we were handling.

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Richard. I took a lot of data this week. I had objects, and I could see them.

If I stitched, depending on how I was walking, and I had to do it on a bunch of data, when I took it out, the image was the same pattern I had stitched. Depending on the object's convexity, when I took out the initial contact points, my gripper closed lower. I could see information that, if I stitched on the same plan, it created a false impression.

I think that's what I want to do. At that point, you just need to create depth information in the training that you like, but we have it because it's the gripper opening. Okay, I understand.

Okay, that's good. Sorry, we can... Okay, I just got what you meant. Because stitching, it was in 2D, but in my mind, we always stitched in 3D the deformation.

It's like if you wanted to do 3D object reconstruction from tactile data. For me, when we talked about stitching, it was that discussion, like we have a tactile map in 2D where we could process it and maybe filter it and get a deformation map in 3D from that, like what I call the tactile imprint, like an impression tactile. Then this can be, if you close a little bit more the gripper, then you can just reference this in the same inertial reference frame and create a shape reconstruction, like popping the object like that.

But what you mean is we're registering tactile maps in 2D, so that's the problem. Yeah, it's not just we register in 2D. We only get the information of the contact point, nothing below, unlike with the image or LIDAR.

Like your LIDAR, you would get the upper point of a ball and the other point, and then if you move your LIDAR or the ball moves, you would get the same profile all along and just move pretty easy to understand the definition of the object. The main point is we only get the same contact as long as we stay on the same reference plane. Like I said, we do know when we change the reference plane because gripper will close more.

Yes, yes, yes. So we know, but the main point I was saying is right now, and I realize it also because Claude, the assumption of the ball was totally right, as long as... So if I take this thing, it's saying outside of when that contact go outside of the sensor, you have no contact anymore, but it's not true. It's going to go to a different layer of the object, and then we'll now generate different contacts.

So the main point is as we don't have, and that could be, by the way, an advantage of a more optical sensor that is translucent. It would be way easier to stitch because you will always get kind of a full picture of, maybe not a full picture, but a good picture of what is around the contact point in terms of depth. So it's easy to stitch, but right now we just get the cross-section, like the point of interception between the object and this.

So I just want to think about that and the way we stitch that because... And again, maybe one solution is just ignoring all this and adds... I still have a hard time to think of a contact where when you switch to a lower map, but the example of the plus sign is one where like moving out of the... But we'll never create an AI that is able to recognize from the first image that it's this object unless we reach this specific object. So I guess it's maybe not that relevant, and we should just... Any time the data force us to change plane, like even if it's in simulation, the ball goes out in a point where the gripper changes plane, we should just stop there and record. Because anyway, due to the complexity, it will happen for a cylinder, it will happen for a ball, but those are always should grasp.

So imagine that there's no grasp to be made there. It's okay in my mind. So it's the point where I say that's the reality that it's a bit more complex to stitch, but we should not handle it in 3D and just keep it in 2D, but taking it as an assumption that whenever like in the data collation, we change plane.

But then you have object where you could have like a wavy object where you always change plane. It doesn't mean that the object is finished there. Okay, why? It's not just like for the ball, it's easy to discard that.

And yesterday, I was taking data with one of Jennifer Fringe object. I don't know if you remember like those object that didn't make to be very work. Yes.

Like the kind of pattern that we're making. But those are very exceptional object. I mean, they are made for that, and none of the object including this old one that we had in the lab made similar like pattern we haven't seen before.

But anyway, just something to think because I remember we said last time that stitching was easy, and we would just take like same thing with anything that is like visual, but it's not totally true here. So that is mine. Besides that, Koush, I wanted to hear about where are you with IDEX and the twin data connection.

So I think we're creating kind of like a very short amount of data with a specific object and just to try it, right? Yes, I'm sorry. I'm not sure I've got the last part you said about the short data. No, no.

I'm just saying that just switching topic from this. Oh, yeah. I think we agreed last time we met a week ago that you would work on trying to just do like a micro pipeline to go to use IDEX and for just a couple of sample.

Yeah, absolutely. Yeah. I want to hear where you are in that.

Yeah, we just wanted to have like a simple plumbing that I can get data automatically for different grids that explore the object. And then I tried stitching them together. I have some results here.

I can share it with you if you don't mind. Yeah. Okay, I'm just going to share my screen here.

Okay, do you have my screen? Yes. Great. So this is, I have it here.

This is the GUI that I made. I can define different grid that I can explore the object with. I can also have my object tilted if I wanted to.

So I, for example, this is one example, one test that I did. I have six points that I explore from the center all the way to the top. And this is how the simulation would look like.

And this is the stitching map that I made. And here, I think this is more clear. This is the blue line is the outline of the cylinder.

And this is the explored area. And after stitching, I get this hot zone here at the center. So I have it for, like, two sensors.

And this is something else. This is the temporal sequential data that I have for the 5%, 50%, and after three seconds of compression. Just curious, Koush.

Yeah. The one you just presented, is color as, like, their... Could you go to the last one? Yeah. Is it possible right now that every of your graph has its own scale, like, to plug the maximum contact to, like, full dark red? Oh.

Because right now, it doesn't make sense to see this evolution of contact here. Right, yeah. No, I have it here.

Would you like to use, like, a fixed map of, like... Sure. No, I have it here for the... See, these are the single ones. And these are all in the same reference.

Like, the color bar range is the same for all of them. So... And then I put them all together and make the big one just for presentation. But this is the 5%, 50%, and this is after, like, the whole pressure.

But, yeah, about the stitching, I have... So this is also another one with a straight line. But I have also a tilted one. Here is another example that I have my cylinder tilted for 20 degrees, I guess.

And this is how the stitching looks like. And the stitching is just the average of the overlapping parts. So I have the big area, and I just populate it with the single tactile data.

And whenever I have overlap, I average them. Could you, in the future, represent this in the way that I represent it with the example that I gave? So what was the original... So what's the original size of the sensor frame? This? To me, it's not clear. Yeah, this is... Unless this is the end of the stitching proposition of everything you took.

Yeah. I'm not sure that I understand. Yeah, this is... See, this is, like... How can I show that? This is the ratio of the cylinder and the sensor pad that I have.

And then I make this movement design. So I have six steps of 6mm in Z. And what I have here is just the whole explored area all together. But, Kourosh... Yeah.

So the sensor is 4x7 taxels. And I tried to count them, but it seems like you have seven increments in the tactile reference frame. So that 1mm would equal one taxel.

Which I find weird a little bit. Do you understand what I mean? It's like you have six increments. And I can see, like, seven shifted six times.

And you average the tactile maps afterwards, if I understood correctly. Yeah, I have... I'm sorry, I'm not sure if I get it right. I have... You said you traveled 1mm at each step? No, no, no, 6mm.

Oh, 6mm at each step? Yeah. And what's the size of one taxel? It's... I have the ratio here. One taxel is 5.5... Okay, okay, okay.

No, no, no. I have a chart that I transfer from taxel to millimeter. And I get this.

This is the real size of the cylinder. And this area is the real size of the explored area. All frames stitched together.

And this is the overlap that I have. Because I'm going from the lower part all the way to the top. And here is the zone that I have the most overlapped taxels.

Okay. But it looks like the top left taxel is always highly activated. Yeah, I thought this... I checked it every other way.

And this is almost true for all of my results. For example, this is a 35 degree tilted. And again, I think this could be coming from the design.

Or... Honestly, I don't have explanation for that. I don't know. Even for the... If we go back to the straight.

Yeah. Yeah, this one is just a simple straight top to bottom. And still here.

I thought maybe... Like as I have... Because in reality it's the same. If I go all the way deep to the object. The sensors are... The sensor pad are not grasping as hard as if I have the contact at the tip.

That was something that happened in the reality. Well, the parallelism of the finger... It really depends on... So when a gripper is new, I forgot which direction. But you will always get a lot of contact.

And when it's old, it's the opposite. You will have a lot of contact at the base. So it could have... It's possible that the model used by your student... I forgot his name.

Encapsulated this. Like it usually shifts around the first 100,000 cycles of a gripper. Okay.

So... Yeah. Can you just show again one tactile map from that? Just without any processing for stitching. Yeah.

Just making sure that this is the data from the correct... This is for the... I think 20 degree tilted cylinder. But if you wanted to... Do you want to see the one related to this? To... Maybe... If... Yeah, sure. No, no, no.

It's just... Yeah, this is the one. So these are the heat maps for the... Yeah. Just... It's all at the center.

Just going up. And the cylinder. So... Yeah.

Like if you... It's my screen. No. Just give me a second, please.

I can just switch because I'm sharing but didn't kick you out. Yeah. No, I stopped.

Now I can see it. So... In this one, I showed the original image. Because at the end, if you want to create a generative AI that uses the extended frame, you will have to train it in a way that here is the original contact you see.

And here is the truth of the extended contact that this represents. The point is, you have to be able to have here with the original image and here is the extended frame. So if you show an extended frame, it's okay.

But I feel like you should show inside. It will really help me to visualize what was your original frame, what was like the initial like you started. And that was the size of it.

Yeah. Yeah. I know the difference is that I... I forced him to move two texels away in all directions.

And like I said, it's not a very good representation, but it's way better. Yeah. But I see, I think that the main difference is that you have one contact and then you have the extended area.

But for me, I have a couple of like a different, like I have six frames stacked together. Yeah. And that shows the... We shifted and that's the whole point of stitching is you know where you move from.

You know you move from like let's say one texel so you can re-center it to where it is. And the point is like if you grasp the cylinder, at some point you will be out of the river and cylinder, you will just get nothing and nothing and nothing. And then if you continue the other side of the cylinder, you will just continue to see things outside of what you initially thought exists.

So... But at the end, it's really what you have to do is to have the original image and then have like a stitched big one of what is the contact outside of the... Like if you would go outside of the tactile map because that's what we want is once you get like with this representation, it would be easier than a garden to move like the grass inside of it thinking that it's just going to be more center, more contact and stuff like that. So that's basically what we want at the end is having this extended. I think it's what you work it with before.

Yeah, yeah. The image you created. So I think it's going to be neat to have at the end.

Yeah, the thing is that for what we're doing now is to get some data so we can use them for the training process. We wanted to train a model. But for data preparation, let's say for now I have six points and stitch them all together and I get one big stitch map.

But for each of them, I have one single tactile that relates to that stitch map. So this is a couple that I'm going to use for the training. So I can plot it that way.

So we have, we can see one single tactile and the stitch area that comes from all those six to, but for, I think I get what you mean. You want to see one single tactile data and the result of the stitch all around it to see how it develops. Extended map.

So if you have a cylinder and then you went higher, let's say you move one tactile each time. Well, you know you move one tactile so you can recenter, you know where your reference frame is. So you can always put the new information in the same reference frame.

Yeah, but the extended area is that only the area that I explored in the Isaac. I don't get any like outside of that. If I go like 20 millimeter.

I understand. You take a first draft on a cylinder and then you go and grab it higher and higher and higher and higher. Exactly.

You get the contact. Like, but again, it's the main point is like, I think we should always start from what was the, because that's the problem we're trying to solve. At the end, we will get an initial contact.

We're not going to explore. So from that contact, what is the truth? Like why I don't know the truth. Maybe we're not going to be able to train it, but you want to be able to try to extrapolate what is outside of it.

Yeah, but which one was by the way, cloud is doing right now, but it's not doing well. So I think we get better when we let's say I have a six different position for the frames. But when I'm using that data to train the model, each of those six position could be the initial position because I'm using all those couples like frame one with the stage frame two with the stage and so on.

So all these could be the initial position because I'm using that for the training. So whenever the model sees the same. What you're saying is kind of true.

We could generate more data this way, but I feel like the problem if we do that, we're not going to have an image that extends symmetrically. Point is, I would probably always give the middle frame and the extended frame around. Okay.

You will have to always extrapolate the same direction. So if you give like the other frame, you capture higher. So to me, the original frame is a year is what you get.

And all the other frame you will move around is to create a larger image where we say here is what exists outside. And we'll create like thousands and thousands of those data using exact sense to have very huge data sets and create generative AI that in the future you will see eight years of contact. And like in my example here, it will generate you like a, instead of a seven by four taxa that would create like a 11 by eight taxa map.

So you will be able to have an idea of what exists around that contact. And the main point is, and it's the reason for doing exact same right now. And I too bad I gave my refer to Jennifer yesterday night, but I could take data for your silencer and stitch them and arrive with this.

But the point is thinking real data will take a lot of times where probably that could create a pipeline with like a hundred of objects. And we just create those data and those data. Because my guess is it will take a lot of data to create the generative AI.

It's not like a classifier this time. Yeah, that's right. Yeah, I get it.

I guess I changed the way I presented and also that was just some new information. Good. I think that I thought I'm going to use all these different position that I get.

So this way I have more data. But if I can also, I can change that. And maybe that's even better to have just the initial data, initial position as the point that I'm using for training.

You could use all those starting points, but then you need to go and extrapolate more. It needs to be like the central frame of what you're going to feed. It's always a different perspective.

It's not really going to work. That's right. Yeah.

And if, I mean, the object is going to end at some point. So if I get like the higher up position as the initial, I don't have enough object to go again and have the same explored area. So if I start at the center.

If you feed different frames that are not middle field, you kind of change the learning problem to an extra variable. It's like, tell me where you are in that frame and extrapolate it in a way that fits. And that will anyway need a lot more data.

So I feel like just keep it in a way that here's the central frame, here's the extended image by two text cells. And yeah, let's just create like a ton of those two images. Right.

No, that makes sense. And I think I have all the parts now to do so. The only, I thought maybe next step is for me to try different objects with different diameter because as I said, for now, for this object, this comes from Barry's example, we have a fixed range of the gripper, like the angle of the gripper that we close around the object for the tactile data to make sense.

So I'm working on... How long does it take right now to take like one data? It takes almost like a two minute for each frame. Two, three, three, three minutes for each. Like if I have a six point grid, it takes 12 minutes.

That's a long time. So is it dependent on the computer where you're working with? Which computer are you... Could be. I'm using the laptop for my laptop.

It's just... And yeah, sometimes it's between like 15 or 18 minutes to 10 or 12. It depends on if I open the Isaac window, like if I run it headless or... But sometimes at the beginning, I want to see what is going on, if I have any. But when I'm sure that it's moving smoothly without any problem or jerky motion, I can do headless.

It's going to save some time. Because two minutes was with the GUI. Yes.

Still open. Okay. Okay, because right now it means that it's 720 data only per... And that is not full frame.

It's just like data, like per day. So let's say you would go on vacation for a week and have your computer. If you only did, it would only be 3,600.

Still faster than... But it should take like six of those images. But right now, six is only one dimension. But you have to move contact in the other dimension.

But Kourosh, I think one thing that we've been doing with Berit is he used Calcul-Québec and also headless, as you already mentioned. I think from what I remember, it could be seconds per frame. Yeah.

My understanding also is shift from physics to Newton. Yeah, we're currently doing that. It seems to work so far.

Yeah, because on our side, for the climatic of the gripper, physics was messing up close to climatic. And now Newton, you know, can really handle it. Okay.

Yeah. So I guess, and it's a lot faster, I think, on our side. So, yeah, it's faster.

So my guess is if you have a version that already works on Newton, you don't want to have noise in what he's doing. So it's not really too interesting. I think he, well, it's really what he's working on right now.

This file is longer than 30 minutes.

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There's probably also if we feel like various pipeline is Improving right now and you work with something in a way that has a lot of issue with changing the object grasping stuff and computing it One way we actually right now to take some basic real data and switching it Because at the end we will probably anyway want to have both of them So I think it's pretty straightforward where you could even just teach pendant the robot It's just like you put an object like it's your cylinder It's fixed and then you move an increment by I don't know five millimeter each time I I think that cell is 5.5 in meters and you can move like by one textile in both directions, so like you to treat axle up to treat axle down and One textile or two on one side of text or two on the other side and you would get that image and it would also be able to get similar thing to validate with this teaching and Right. Yeah. No, sure.

I'm gonna answer like Okay, I we can talk about this on Friday right we if I can Thank you Thank you so much on my side Sorry, just I'm gonna be in Europe. So what time will it be? I cannot yeah, I think five or Okay, you're fine. So yeah 11 a.m. August 4 Perfect.

Thank you so much Thank you for a time Vinson John Flipp, yeah, and I'll see you John Flipp on Friday Okay, it works. Thank you so much. Have a great day.

Thanks. Bye You